

Design clustering of offshore wind turbines using probabilistic fatigue load estimation



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ABSTRACT

In large offshore wind farms fatigue loads on support structures can vary significantly due to differences and uncertainties in site conditions, making it necessary to optimize design clustering. An efficient probabilistic fatigue load estimation method for monopile foundations was implemented using Monte-Carlo simulations. Verification of frequency domain analysis for wave loads and scaling approaches for wind loads with time domain aero-elastic simulations lead to 95% accuracy on equivalent bending moments at mudline and tower bottom. The computational speed is in the order of 100 times faster than typical time domain tools. The model is applied to calculate location specific fatigue loads that can be used in deterministic and probabilistic design clustering. Results for an example wind farm with 150 turbines in 30–40 m water depth show a maximum load difference of 25%. Smart clustering using discrete optimization algorithms leads to a design load reduction of up to 13% compared to designs based on only the highest loaded turbine position. The proposed tool improves industry-standard clustering and provides a basis for design optimization and uncertainty analysis in large wind farms.

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1. Introduction

The offshore wind energy industry faces three important trends: (1) wind farms grow in size, (2) monopiles are installed in deeper water, and (3) cost reduction remains the most important challenge [1]. Part of the cost reduction must be realized from optimization of support structures for offshore wind turbines (OWTs). The governing support structures are monopiles as they cover 75% of the offshore wind market in 2013 [2]. Large offshore wind farms are predominantly located in deeper water (20–40 m) covering areas of several square kilometers (e.g. DanTysk with 80 OWTs (3.6 MW) covers ca. 70 km²). Therefore, environmental site conditions, such as water depth and soil conditions, can easily differ significantly from one turbine position to another within large offshore wind farms. These variations lead to significant differences in fatigue loads on OWT support structures. Ideally, support structures should be optimized individually for each turbine position. In industry practice however, loads are often only evaluated for a limited number of design positions in the wind farm since load calculations

are computationally demanding and cannot be performed for each position [3]. OWTs are then grouped into design clusters according to the design positions. For each cluster the preliminary support structure design is only performed once. It is assumed that the resulting load levels and structural designs can safely be carried over to all positions in the associated cluster. In order for this to hold, the design position must be the highest loaded location in the cluster. Thus, optimization of design clustering is necessary to reduce design conservatism and the cost of offshore wind energy.

Allocating OWTs to design clusters is typically performed in early project phases (preliminary design), making time-efficient approaches essential while load estimates are sufficient. In current industry projects turbines that are positioned in similar water depth and/or are expected to have similar first eigenfrequencies of the support structure are grouped into clusters [4]. Recently, Seidel [4] suggested an improved approach of clustering using a site parameter that also takes into account structural and hydrodynamic properties. This site parameter is however only suitable for wave load dominated designs. Thus, further work is needed to formulate turbine clustering as an optimization problem incorporating all important site specific information.

In detailed design and certification of OWT support structures a high number of load cases have to be assessed, which makes the

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computational effort for standard time domain fatigue load analysis significant [5,6]. Such simulations, although they are considered most accurate, are ill-suited for clustering optimization and probabilistic assessments. For these applications, load estimates are sufficient, as fast simulations are a necessity. Several approaches exist in literature to decrease computational costs of OWT load analysis, either using integrated analysis or wind-wave separate assessment.

A separate assessment of aerodynamic loads from time domain simulations combined with calculation of hydrodynamic loads in the frequency domain is suggested by Kuehn [7]. Later, van der Tempel [8] extends frequency domain analysis to wind-induced fatigue load calculation emphasizing the need to account for aerodynamic damping as the most important dynamic interaction [9]. In real systems, coupled wind and wave excitation leads to dynamic interaction between aerodynamic loads from the turbine rotor and hydrodynamic loads on the support structure. Aerodynamic damping describes an aerodynamic force that opposes the direction of turbine movement caused by changes in relative wind velocity on the blades due to tower movement in the wind direction (fore-aft mode) [10]. Ragan and Manuel [11] emphasize that spectra of structural response of wind turbines are often not narrow-banded and Gaussian due to rotor periodicity, hence stress ranges do not follow a standard Rayleigh distribution. They suggest to use the empirical Dirlik's method [12] to obtain stress range distributions from wind-induced load response spectra. In integrated methods, researchers propose to reduce the number of environmental conditions [13], number of simulation seeds, or simulation length [7]. For example, Zwick and Muskulus [13] suggest simplified fatigue load assessment based on statistical regression models to reduce the number of load cases for damage estimation.

Fatigue load calculations contain significant uncertainties commonly addressed with safety factors in the design standards [6,14]. Insight on the effect of uncertainties on EFLs improves the understanding of the actual structural reliability. A brief overview of existing work on probabilistic fatigue assessment of wind turbines is given by Yeter [15] and Veldkamp [16].

In this paper, a design clustering method for OWTs was developed using probabilistic fatigue load estimation. The fatigue load estimation method uses frequency domain analysis to calculate wave loads and a scaling approach for wind loads on monopile-based OWTs. Uncertainties in input data and model assumptions are analyzed through Monte-Carlo simulations (MCS). The focus is on uncertainties in the load calculation model, while structural response and damage is determined deterministically. The load estimation tool was verified with fully integrated, aero-elastic time domain simulations. Due to computational efficiency, fatigue loads can be calculated site specific for every turbine location within the wind farm. This makes turbine clustering a discrete optimization problem.

This paper provides a novel approach of optimization of turbine clustering. Additionally, probabilistic assessments give insight on how uncertainties in load calculation can alter allocation of turbines to clusters. This paper is organized as follows. Section 2 outlines the developed fatigue load estimation method, its model assumptions and verification. A summary of the probabilistic load assessment is presented in Section 3. Section 4 focuses on clustering optimization by discussing problem formulations and solution approaches using brute-force and discrete optimization algorithms. Section 5 demonstrates the established clustering optimization method for a realistic wind farm example of 150 monopile-based OWTs in a water depth range of 30–40 m exposed to typical North Sea conditions.

2. Method for fatigue load estimation

2.1. Approach

The developed method calculates dynamic structural response of OWT support structures to site specific aero- and hydrodynamic loading. Fatigue loads for shear forces and bending moments are estimated for distinct output locations of the support structure (mudline and interface) for each turbine position in the wind farm. The term “interface” refers to a structural node between tower bottom and top of the monopile. Loads are assessed as equivalent fatigue load (EFL) which is a constant-amplitude load range that would cause an equivalent amount of damage as a variable-amplitude load time series F_i for a specified number of load cycles N_k (cf. Equation (1)). N_i are the number of cycles, T_{sim} is the simulation time, T_{life} is the lifetime, and m is a material parameter from the SN-curve of welded steel.

$$EFL = \left(\sum_{i=1}^n \frac{N_i}{N_k} \cdot \frac{T_{life}}{T_{sim}} \cdot F_i^m \right)^{1/m} \quad (1)$$

The method is based on three core elements (cf. Fig. 1): (1) calculation of wave-induced EFLs in the frequency domain, (2) scaling of wind-induced EFLs from a time domain reference case, and (3) combination of wind-wave EFLs with direct quadratic superposition [7]. The execution time for one simulation case is in the order of seconds.

Wave-induced fatigue loads: The estimation of wave-induced fatigue loads in the frequency domain is summarized in the following. Full details of the method are given in Ref. [17]. Hydrodynamic loads are obtained from Morison equation using wave kinematics power spectral densities based on JONSWAP wave spectra. The drag force in Morison equation is linearized by using the first term of a Fourier series expansion as suggested by Borgman [18]. Modal analysis is performed on a finite element model of the OWT support structure to synthesize transfer functions using the modes in the frequency range of interest at the desired input and output degrees of freedom.

To account for the dynamic interaction between wind and waves, aerodynamic damping (damping ratio ca. 1.5–8%) is superimposed to structural, hydrodynamic and soil damping (damping ratio ca. 1%) in the synthesis of transfer functions. Aerodynamic damping is a function of wind speed, rotor speed, and mode shape. It is determined turbine-specific with modal analysis from a fully coupled, non-linear aero-elastic model. The aerodynamic damping force acts in opposite direction to wave excitation forces on the support structure in case that wind and waves are unidirectional (assuming no yaw misalignment). Therefore, aerodynamic damping is largest for the first mode (fore-aft) and aligned wind and waves, while the damping effect reduces for wind-wave misalignment and higher structural modes.

The finite element model describes foundation and tower with Timoshenko beam elements, rotor-nacelle-assembly as an equivalent concentrated mass on top of the tower, and soil with distributed linear springs according to a realistic reference design (further details are given in Ref. [17]). In order to minimize computational effort, transfer functions are generated for a moment-equivalent wave force at MSL only instead of using distributed transfer functions along the monopile wave-action zone. The distributed wave loads are integrated to MSL and weighted with the first mode shape in order to avoid overestimation of the equivalent load. Structural response is determined by combining transfer functions with input hydrodynamic loading. Wave-induced EFLs are finally obtained from the response spectra with Dirlik's method [12].

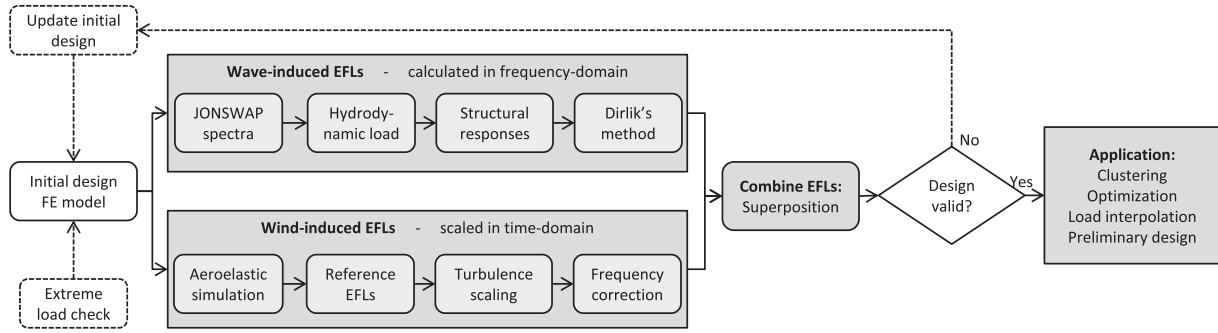


Fig. 1. Flowchart of fatigue load estimation method.

Wind-induced fatigue loads: Wind-induced fatigue loads are estimated with a time domain scaling method. In this method, full aero-elastic simulations of aerodynamic loads on an OWT in a reference setup are run with a non-linear time domain tool for global dynamic analysis of wind turbines, Bonus Horizontal axis wind turbine Code (BHawC). BHawC uses Blade Element Momentum Theory to calculate lift and drag forces on each blade element with an expansion for skewed and unsteady inflows, correction for high induction values and dynamic stall, and Prandtl's tip loss correction [19]. In the reference setup all fatigue relevant load cases are simulated for a specific wind turbine (10-min simulations). Resulting structural response time series are post-processed with rainflow counting to obtain the number of cycles per load range. For each simulation, the variable-amplitude load ranges are converted to EFLs which now form the reference case $EFL_{wind,ref}$. This reference case is turbine type specific, but assumed generic regarding environmental conditions and support structure dimensions.

The reference case $EFL_{wind,ref}$ for OWTs consists typically of Class C turbulence intensity TI_{ref} [6]. Location specific wind loads $EFL_{wind,ls}$ are then approximated by applying linear scaling of the turbulence intensities and a natural frequency correction factor, f_{corr} , as shown in Equation (2). The location specific turbulence intensities, TI_{ls} , are calculated with the Frandsen wake model using input from measured, ambient TI [20]. The natural frequency correction is derived from interpolation of time domain simulations for four different turbines types with 3–7 frequency sets each.

$$EFL_{wind,ls} \approx EFL_{wind,ref} \cdot \frac{TI_{ls}}{TI_{ref}} \cdot f_{corr} \quad (2)$$

Combination of wind- and wave-induced EFLs: EFLs from wind and waves are combined using Equation (3) [7]. Direct quadratic superposition is accurate for narrow-banded response spectra at the same frequency (identical zero-crossing period of wind- and wave-induced responses) [7]. Equation (3) gives conservative results for broad-banded response spectra since the neglected interference term is commonly assumed to be negative [21]. Alternatively, summation of response spectra [7] and applying Dirlik's method on the combined spectrum can reduce errors in EFL estimation for broad-banded response superposition.

$$EFL_{tot} \approx \sqrt{EFL_{wind,ls}^2 + EFL_{wave,ls}^2} \quad (3)$$

Direct quadratic superposition of EFLs is chosen in this approach because of the statistical model for estimation of location-specific wind-induced EFLs (cf. Equation (2)). The turbulence scaling and natural frequency correction is only feasible using EFLs but not with response spectra. Additionally, the errors induced by Equation (3) are minor if one of the load components is dominating. This

makes the method suitable for designs governed by wave-induced fatigue loads, which is often the case for large-diameter monopiles in deeper water.

2.2. Assumptions

To create a computationally efficient method, several assumptions are implemented to simplify full load calculations. Detailed assumptions of the wave load estimation are outlined in Ref. [17]. Additionally, the following assumptions are important:

2.2.1. Assumptions for environmental data

Wind-wave scatter diagrams are lumped into 20 simulation cases consisting of mean wind speeds (V_W), significant wave heights (H_S), and wave spectral peak periods (T_P). The lumping is performed in a damage equivalent way using a method suggested by Kuehn [7]. For each simulation case wind-wave directionality is simplified into fully aligned or fully misaligned. For this purpose, wind and wave roses are lumped in bins of 30°. Fully aligned wind and waves occur in the same bin, where the aerodynamic damping is maximum. All other combination of wind and wave directions are treated as fully misaligned with reduced aerodynamic damping.

2.2.2. Assumptions for calculation method

The estimation approach for wind-induced EFLs only adjusts for TI and natural frequency. Differences in air density, wind shear, structural geometry or mode shapes are neglected in the estimation of structural response from aerodynamic loading since the effects on the load level are expected to either be minor or represented through the natural frequency correction. The finite element model used for calculation of wave-induced EFLs, however, is adjusted for site-specific water depth. In industry practice, individual structural designs are made in the detailed design phase. Beforehand, turbine clustering is determined in the preliminary design phase based on a generic structural design due to simplicity assuming that clustering results do not change significantly for design adjustments.

2.3. Verification

The fatigue load estimation method is verified empirically against time domain simulations using the non-linear aero-elastic tool BHawC. The verification was done for wave-only load cases (as described in Ref. [17]), as well as wind-only loads, and wind-wave combined loads for a 4 MW OWT in 35 m water depth. Results of this verification study are presented in Section 5.

Table 1

Selected sources of uncertainty in case study for probabilistic design clustering. Mean values of the random variables are based on data of the example wind farm. Standard deviations are derived either from wind farm data (W), references, or expert opinion (E).

Parameter	Mean	STD	Ref.
Data (factors)			
Turbulence intensity [-]	1	0.1	[16]
Mean sea level [-]	1	0.05	W
Soil stiffness [-]	1	0.2	[23,17]
Statistical uncertainty			
Lumped H_s [m]	1.6	0.1	E
Lumped T_p [s]	6.9	0.2	E
Model uncertainty			
γ [-]	1.5	0.1	[22]
C_M [-]	1.7	0.1	[16]

3. Probabilistic load assessment

Epistemic uncertainties in fatigue load calculations occur in the form of data, statistical and model uncertainty [22]. Previous work showed significant influence of data uncertainties in MSL and T_p on fatigue loads [17]. For the clustering problem, the probabilistic assessment is extended to include uncertainty in TI and model uncertainty. A large number of important model uncertainties exist in the complex analysis of OWTs and their environment. This study accounts only for two selected model uncertainties in the wave load model but does not consider uncertainties in structural response and damage models. Load model uncertainty is analyzed regarding the development of sea states, represented with the peak enhancement factor γ in the JONSWAP wave spectrum, and regarding diffraction, represented with the inertia coefficient C_M in the McCamy-Fuchs diffraction correction for Morison equation.

The probabilistic load assessment and clustering is demonstrated in a case study with a realistic wind farm example. The example wind farm consists of 150 4 MW OWTs with monopile support structure. It is located in typical North Sea conditions as presented in Table 2. For every turbine location 1000 MCS were run with simultaneous, random variation of input parameters. As a result, the output distribution of EFLs can be estimated with a root-mean-square error of 0.3%.

Random variables in the case study for probabilistic load assessment are independent, normal distributed with mean values and standard deviations (STD) stated in Table 1. The mean values of input parameters in Table 1 were chosen based on real wind farm data to represent realistic load uncertainties for design clustering in large wind farms. Mean values of data factors are normalized since exact input values cannot be disclosed. However, upper and lower bounds of the site-specific input variables are given in Table 2: water depths cover 30–40 m, TI range between 0.10 and 0.19 and are calculated with a Frandsen wake model [20], and soil consists of mostly dense sand with scarce clay layers. Mean values of γ and C_M are chosen based on wind farm and support structure data about fetch limitation and diffraction (cf. Table 1). Probabilistic clustering is demonstrated using only one lumped fatigue sea state of

$H_s = 1.6\text{ m}$ and $T_p = 6.9\text{ s}$ in order to reduce computational effort. In practice, however, detailed support structure design must account for all fatigue relevant load cases.

STDs of the random variables are determined based on wind farm data, existing publications of sensitivity studies [16,17,22,23], and expert opinion (cf. Ref. in Table 1). For example, the variation of MSL is chosen as tidal variation from the metocean data of the example site. Soil properties typically contain high uncertainties due to difficult measurement and data interpretation while the water depth is known with better accuracy. Soil variations are represented by scaling of the soil stiffness which is obtained from nominal p-y curves with a factor that is constant over the full depth. The variability of lumped H_s and T_p presents only the uncertainty in damage-equivalent lumping of sea states (cf. Chapter 2.2) and is therefore smaller than the complete long-term sea state variability.

Mean values are location specific for data uncertainty, while they are constant over the wind farm with respect to statistical and model uncertainty. STDs are identical for all turbine positions. STDs for data uncertainty are represented as a scaling factor. In practice, uncertainties can differ for some locations due to incomplete measurement data, for example soil measurements may not be available for every turbine position in early project phases.

4. Clustering optimization

The aim of design clustering is to find an optimum between number of clusters and the level of design conservatism. Clustering optimization here provides a solution to the question of cluster configurations: To which cluster should each OWT be assigned to?

Applying the developed (probabilistic) estimation method, fatigue loads are calculated site specific for all given turbine locations within the wind farm. The resulting set of discrete (deterministic or probabilistic) fatigue values is clustered regarding customized criteria.

4.1. Problem formulation

Clustering optimization has two dimensions: the decision on (i) number of clusters and (ii) cluster configuration. Regarding number of clusters, limiting cases are one cluster, meaning all turbines are designed for the highest load level in the wind farm, and individual support structure design, where cluster number equals number of OWTs. The number of clusters is typically restricted by resources in the engineering team and limited project time. Even if the industry moves towards faster tools and higher capacities making individual design feasible, clustering is still beneficial to implement for turbines with similar load level to optimize costs of manufacturing and installation.

Ideally, in clustering optimization the support structure costs should be minimized. In this study, the optimization problem is formulated to minimize design loads since accurate cost information is difficult to gather. Loads correlate to mass of the support structure which is commonly treated as cost indicator [7]. Thus, the unconstrained, non-linear objective function is the total design load TL calculated as the sum over all clusters n with Equation (4). N_i is the number of turbines in cluster i , while L_i equals the highest load level in cluster i .

$$TL = \sum_{i=1}^n N_i \cdot L_i \quad (4)$$

The choice of number of clusters n can either be fixed prior to optimization or be included as a variable in optimization. If the clustering approach is extended for cost minimization, n will be an optimization result. Including uncertainties in fatigue load

Table 2

Data of example wind farm.

Rated power	4 MW
Support structure	Monopile
Location	North Sea
Water depth	30 - 40m
Turbulence intensity	0.10 - 0.19
Soil conditions	Dense sand

calculations results in a set of fatigue load distributions for all turbines. Simplified probabilistic clustering is performed by increasing the mean load value with $k \cdot \text{STD}$. Choice of k represents how much of the load variability is accounted for, for example $k = 3$ accounts for 99.7% of the sample population for a normal distribution. For smaller values of k more influence is given to the mean value of EFLs, while for larger k values STDs become more important. Consequently, the higher k is chosen, the smaller is the chance of misgrouping turbines into clusters with too low load level. As an alternative, full probabilistic clustering can be based on a distribution of the cluster load level L_i which leads to a probability distribution of TL for given N_i . Optimization can then be performed by minimizing a chosen characteristic value of TL . This will only lead to different results compared to the simplified probabilistic clustering if Equation (4) is non-linear with respect to L_i , for example once support structure costs are included. Problematic in this approach is setting up a representative distribution of L_i based on individual EFL distributions of each turbine in the cluster. Therefore, this is left to future work.

4.2. Assumptions for clustering optimization

The formulation of the optimization problem reveals underlying assumptions for clustering:

- All turbines are designed for the highest load in the cluster, no design interpolation is done within clusters.
- Fatigue is design driving, extreme loads do not influence clustering. Designs are often dominated by either wind- or wave-induced fatigue loads depending on water depth, wind/wave conditions, support structure type and wind turbine size. Wave-induced fatigue loads typically play a major role for large diameter monopiles in deeper water. A check should be performed for every project to confirm fatigue as a design driver.
- Location specific loads are calculated with the same initial design for all positions. Thus, the validity of the design for all positions has to be confirmed which can be done through the frequency constraint. Alternatively, the initial design can be updated in an iterative procedure of clustering and renewed load calculation.

4.3. Solution approaches

The pattern of the optimization problem is explained for an example of 150 turbines equally divided into five clusters with cluster 1 ($T_1 - T_{30}$) containing the highest loaded turbines and cluster 5 the lowest loaded turbines ($T_{120} - T_{150}$). Thus, 30 turbines are in each cluster ($N = 30$). Moving one turbine T_{60} from cluster n_2 to cluster n_3 causes:

1. $(N - 1)$ turbines in cluster n_2 ,
2. $(N + 1)$ turbines in cluster n_3 and
3. Increase of maximum load in cluster n_3 to L_{60}

The formulated problem is addressable with algorithms of combinatorial optimization [24]. The discrete solution approaches brute-force and local search were compared in terms of efficiency and robustness. In **brute-force** the number of recombination are reduced by using pre-knowledge of the solution that (i) clusters are filled with turbines sorted in load descending order making optimization only a cluster-border decision and (ii) when adding a cluster, the number of turbines in every cluster is less or equal to the maximum number of turbines in the previous n clusters. Problems with 150 OWTs and up to 6 clusters can be solved with

brute-force giving the global minimum in a computation time of less than 1 h.

An approximate algorithm using **local search** starts from an initial cluster guess, selects a random cluster-border, and optimizes it in terms of decreasing the objective function, while keeping all remaining positions constant. The optimization of the single cluster-border is performed by selecting the position with the minimum cost of all possible border positions. This procedure is repeated in every iteration step. With a chance of 5%, the previously optimized cluster-border is replaced by a random border to ensure that the algorithm does not become trapped in a local minimum with poor quality. In the tested example for 150 OWTs, 100 iterations were sufficient for the algorithm to reach the global minimum or local minima with good quality. The quality of local minima depends on the shape of the solution space and the initial guess. In order to avoid poor local minima, the optimization algorithm is repeated for several initial positions selected randomly. In a fast check, the quality of the computed local minima is compared against a linear distribution of load levels in the offshore wind farm and results for individual design. For a linear distribution of loads, the optimal solution is an equal number of turbines in each cluster.

5. Results and discussion

5.1. Probabilistic fatigue load estimation results

Results of the verification study for the load estimation method are presented in Fig. 2. Table 3 states the input sea states for the verification study and deterministic clustering. For each operational wind speed from 4 m/s to 25 m/s one damage equivalent sea state is lumped from long-term wind-wave scatter diagrams. The values are presented in groups. In the simulation, one distinct $H_S - T_p$ combination was used for each operational wind speed. In the time domain simulations with BHawC seven random realizations (seeds) were used for every wind speed and corresponding sea state to minimize statistical deviations. The load estimation tool gives the best results around rated wind speed of the turbine. Errors increase for higher wind speeds mainly due to an underestimation of wave loads. This is in agreement with previous study of the wave-only load estimation tool [17]. Possible reasons are that for higher wind and waves non-linear effects in the wave load and also in the soil-structure interaction play a larger role which is not captured in the linear frequency domain wave model. Additionally, aerodynamic damping magnitude, small differences in the calculation models (e.g. discretization in structural model, C_M coefficients in Morison equation), and response combination with

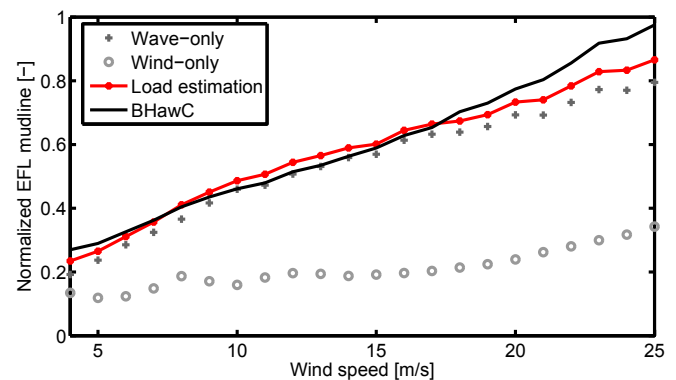


Fig. 2. Comparison of normalized EFLs of response moment at mudline from the load estimation tool with time domain aero-elastic simulations using BHawC. The shares of wave-only and wind-only EFLs from the load estimation tool are included.

Table 3

Lumped sea states corresponding to all operational wind speeds.

V_W [$\frac{m}{s}$]	H_S [m]	T_p [s]
4	0.0–0.5	4.0–4.5
5–8	0.5–1.0	4.0–4.5
9–10	1.0–1.5	4.5–5.0
11–13	1.5–2.0	5.5–6.0
14–15	2.0–2.5	6.0–6.5
16–17	2.5–3.0	6.5–7.0
18–19	3.0–3.5	7.0–7.5
20	3.5–4.0	7.5–8.0
21–22	4.0–4.5	8.0–8.5
23–25	4.5–5.0	8.5–9.0

direct quadratic superposition are further error sources. Combining the simulation cases to weighted lifetime EFLs using a Weibull distribution for wind speed occurrences, a total load estimation accuracy of 95% (conservative error, EFLs are overestimated) is achieved.

Fig. 3a presents the location specific EFLs for 150 turbines calculated with the load estimation tool for a realistic wind farm example in 30–40 m water depth leading to a maximum load difference of 25% in the wind farm. The mean values of MSL, soil properties, and TI are site specific, while all remaining parameters are constant, e.g. identical lumped sea states or wind-wave alignment. Sorting the turbines for ascending mudline EFLs does not result in interface loads sorted correctly. This is due to the higher local sensitivity of wave-induced EFLs at mudline nodes to changes in MSL as shown in Ref. [17]. Thus, a critical decision in clustering optimization is the position of interest, since clustering regarding mudline EFLs might result in interface EFLs being in a suboptimal cluster. In this study, optimization of mudline EFLs was performed since these are typically the highest loaded nodes on the support structure. Alternatively, multi-objective optimization or an optimization based on the “integrated load from interface-mudline-pile tip” might be suitable.

Fig. 3b shows the normalized mean load values with an error bar for STDs of all turbines resulting from the probabilistic load assessment. Mean load values are normalized to the maximum mean load value in the wind farm. In general, STDs between the turbines are quite similar; the maximum absolute difference is 4.5%.

A reason for the relatively small variation between location specific STDs of EFLs is the choice of input distributions. In this example case, only the data factors for MSL, soil, and TI introduce location specific uncertainties, while the remaining uncertain

parameters are identical for all turbines (cf. Table 1). Scatter plots of EFLs for the input parameters reveal that uncertainty in MSL, soil stiffness, T_p , H_S , and C_M influence the loads equally while γ and TI have no relevant influence (not shown). Higher location specific uncertainties may increase variability differences within wind farms.

Fig. 4 presents STDs of EFL distributions of all 150 turbines plotted as a function of location-specific natural frequency, MSL, and normalized load level. The following two trends regarding the load variability are observed:

1. STDs of EFLs increase with higher load level and accordingly also with deeper water or lower eigenfrequency. This result matches previous analysis that showed a higher sensitivity of EFLs for increasing water depth and decreasing soil stiffness [17].
2. For turbines on a similar load level, differences in STD up to 2% exist. This variability can be attributed mainly to location specific uncertainties since the effect of model and statistical uncertainty is constant for the same load level.

5.2. Clustering results

Clustering based on only eigenfrequency and MSL gives an indication but does not fully capture the variation of EFLs as Fig. 5a shows. The representation of EFL change through eigenfrequencies is of higher quality than for MSL. A better match was achieved by Seidel [4] with his suggested load site parameter. Using the developed load estimation tool, clustering can now be performed based on complete site specific fatigue load information.

Fig. 5b shows the total design loads (TLs) of all possible re-combinations for grouping 150 turbines into three clusters using brute-force. TLs are presented as a function of number of turbines in cluster 1 (highest loaded group) and cluster 2. The remaining cluster 3 (lowest loaded group) contains the number of turbines that were not yet allocated to cluster 1 or 2 ($N_3 = 150 - N_1 - N_2$). The solution space of the optimization problem has a convex pattern with a rather flat bottom with local minima that only differ marginal from the global minimum (cf. Fig. 5b). TLs increase at the cluster margins (three peaks), where the majority of turbines is allocated to a single cluster causing many turbines to be over-designed. This shape and the relative smoothness of the problem enable local search algorithms to achieve good results.

The local search algorithm for optimizing 10 clusters is applied for 20 random initial positions as shown in Fig. 6a. This confirms that the local search algorithm leads to a very good local minimum

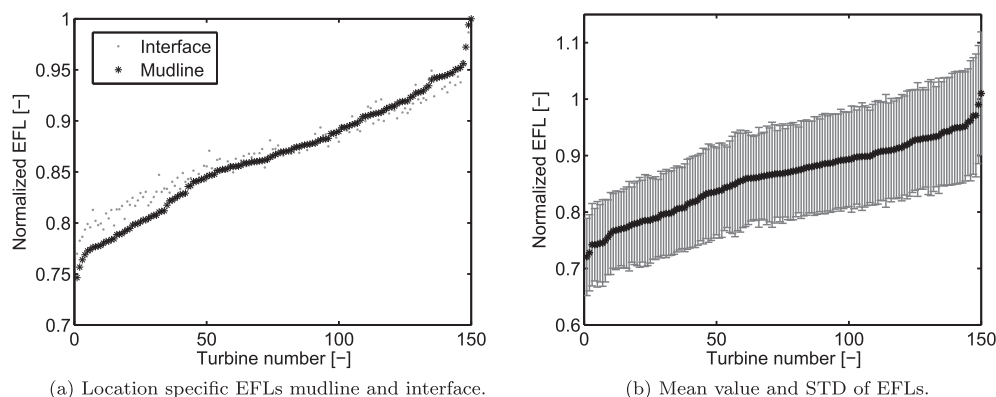


Fig. 3. (a) Location specific EFLs for bending moments at mudline and interface level for 150 turbines normalized to the maximum load in the wind farm. The EFLs are sorted ascending for mudline location. (b) Mean value and STD of probabilistic, location specific EFLs for bending moments at mudline.

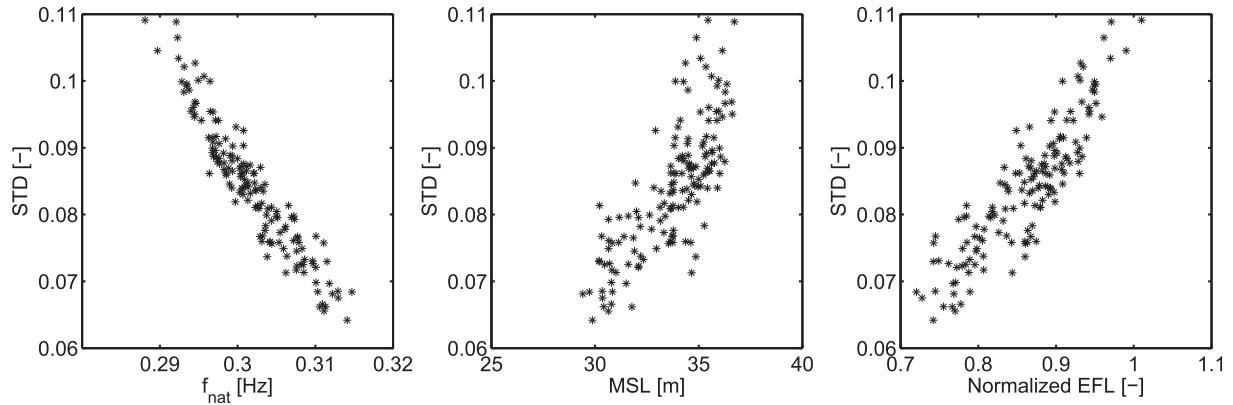


Fig. 4. STDs of EFL distributions for 150 turbine positions with location-specific site conditions plotted as a function of natural frequency, MSL, and normalized EFLs.

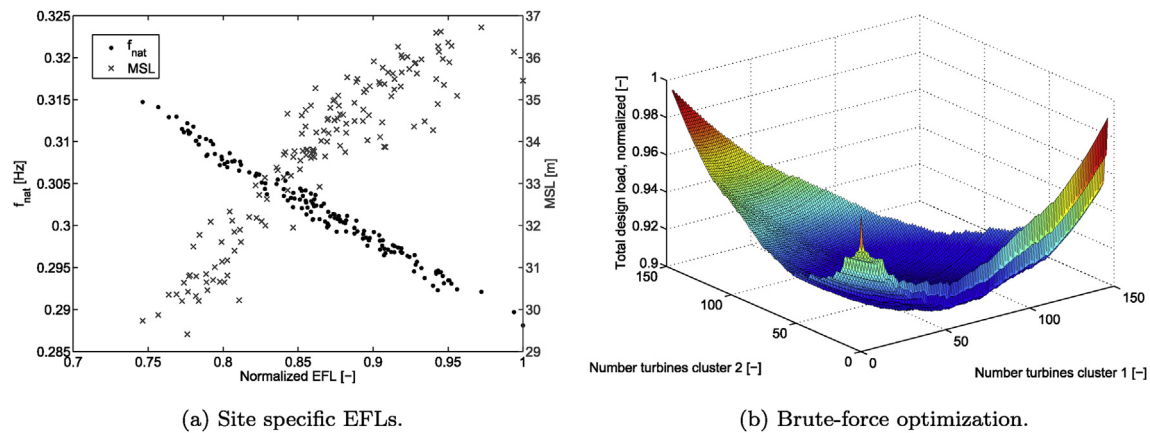


Fig. 5. (a) Scatter plots of site specific EFLs for eigenfrequencies of the structures f_{nat} and MSL variation. (b) Brute-Force optimization of 150 turbines in 3 clusters.

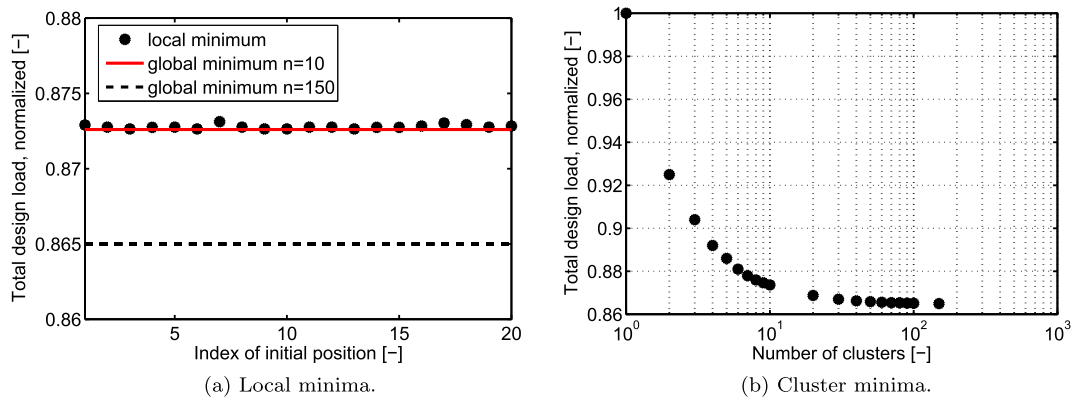


Fig. 6. (a) Local minima obtained with discrete optimization algorithm for different random initial positions. (b) Minima of TLs for 150 turbines as a function of number of clusters.

with only minor influence of initial positions. The maximum difference between the different local minima is 0.4% for this example which is negligible.

Fig. 6b presents TLs minima as a function of the number of clusters. The clusters 1–6 are calculated with the brute-force method resulting in the global minimum, while higher number of clusters are solved with the local search algorithm. One cluster means all turbines are designed for the highest load in this wind farm, while 150 clusters represent individual design. This emphasizes the necessity to implement a sufficient number of clusters in

large offshore wind farm since designing all support structures to the highest loaded position leads to 13% larger TLs compared to individual design for this example case. On the contrary, the marginal gain adding more clusters reduces significantly after 6 clusters. TLs approach individual design results quickly after 10 clusters already, where more designs add little value.

The optimal cluster configurations are shown in Table 4. Optimization has to be performed for the number of turbines in each cluster N_i only; the maximum load in the cluster L_i results from ascending turbine sorting (cf. Fig. 3a). This setup based on brute-

Table 4
Optimal cluster configurations.

Number of clusters		1	2	3	4	5	6	150
Cluster 1	N_1	150	54	24	18	4	4	1
	L_1	1	1	1	1	1	1	1
Cluster 2	N_2	—	96	53	41	30	20	1
	L_2	—	0.883	0.919	0.929	0.952	0.952	0.994
Cluster 3	N_3	—	—	73	57	43	30	1
	L_3	—	—	0.862	0.878	0.908	0.919	0.972
Cluster 4	N_4	—	—	—	34	39	25	1
	L_4	—	—	—	0.812	0.862	0.883	0.956
Cluster 5	N_5	—	—	—	—	34	37	1
	L_5	—	—	—	—	0.812	0.861	0.952
Cluster 6	N_6	—	—	—	—	—	34	1
	L_6	—	—	—	—	—	0.812	0.951
Cluster 150	N_{150}	—	—	—	—	—	—	1
	L_{150}	—	—	—	—	—	—	0.746
Normalized TL		1	0.925	0.904	0.892	0.886	0.881	0.865

force solving can also be used for interpolation to higher number of clusters instead of applying optimization algorithms.

Next, based on the calculated load distributions (cf. Fig. 3b), clustering is now performed probabilistically by allocating turbines based on their mean load level increased with $3 \times \text{STD}$. This is presented for an example of three clusters in Fig. 7 where the rearrangement of turbines in probabilistic clustering is compared to results from deterministic clustering based on only mean values. In this example, the number of turbines in cluster 1 increased by one while cluster 3 has one turbine less. In Fig. 7a, the numbers indicate turbine IDs, grey stating the turbines in the cluster for deterministic clustering and the red boxes referring to turbine moving due to probabilistic clustering. In total, twelve turbines change clusters as indicated by the arrows. These are often turbines located close to the cluster-borders but can also be turbines that have a higher STD than load neighbouring turbines (e.g. ID23). This demonstrates that probabilistic clustering can cause differences compared to deterministic clustering due to distinct dynamic properties of turbines caused by individual soil and water depth situations.

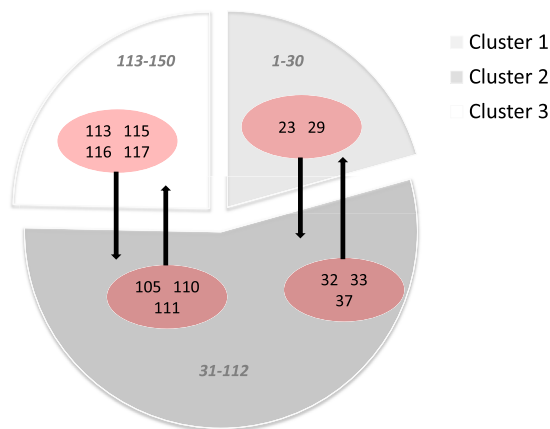
Turbines changing clusters leads to an increase of the highest mean EFL in the clusters. Therefore, the total design loads are higher for probabilistic clustering compared to deterministic clustering as shown in Fig. 7b.

5.3. Evaluation of load estimation and clustering methods

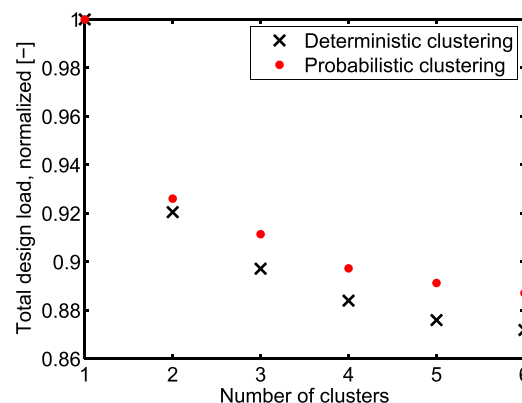
Limitations of the current estimation tools are the simplification of many physical phenomena, for example non-linearity of waves and soil, wind-wave directionality and interaction between wind and wave loads. Moving to bigger turbines supported by XL-monopiles with diameters above 7 m, wave diffraction becomes more important, making use of Morison equation in linear wave theory only applicable when diffraction corrections are included. Immediate improvement of the method can be achieved through the use of distributed transfer functions in the wave load tool, detailed study of suitable aerodynamic damping values, and replacement of direct quadratic superposition by a summation of wind- and wave-induced response spectra. Verification of the fatigue estimation method confirms a total conservative error of 5% indicating good load estimation. However, since some results of the estimation tool are non-conservative for high wind speeds (cf. Fig. 2), contingency should be added in project applications.

Probabilistic assessment of all turbines within an example wind farm gave only 5% differences in STD over the wind farm. For design practice, it is suggested to perform a single probabilistic assessment for the highest loaded position in uniform wind farms to indicate the variability of loads due to uncertain input parameters for all turbine locations. With some contingency, this approach is conservative as it was shown that STDs increase with higher loads.

A main limitation of the probabilistic assessment is that results



(a) Turbines changing clusters.



(b) Design loads.

Fig. 7. (a) Turbines changing clusters in probabilistic compared to deterministic optimization. (b) Total design loads for deterministic and probabilistic clustering.

strongly depend on chosen input distributions. For example, soil stiffness had only minor influence on EFLs compared to parameters like MSL and T_p in previous studies [17]. This was due to high soil stiffness for the example site mainly consisting of dense sand (North Sea location) and scarce clay layers. Soil stiffness became in this study a more important parameter for turbine locations with poorer soil stiffness. This emphasizes that probabilistic assessments are problem specific and cannot be generalized. Consequently, applying a general safety factor in the design process is not expedient for design optimization. Probabilistic load assessment can, combined with uncertainty in structural resistance, form a good decision-making basis for optimizing safety factors. For design clustering, the needs of industry are fast and robust tools, while nowadays larger wind farms typically have only 2–3 clusters. Brute-force optimization guarantees global optima at the expense of solution time making it the best choice for up to 5 clusters for wind farms with 100–150 turbines. For more clusters, the local search algorithm is recommended which showed very good performance for solution quality and computational efficiency. For practice, a possible constraint in the optimization routine can be a limit on MSL difference within clusters so that monopile lengths can be kept constant for all cluster positions. In the probabilistic clustering example, only few turbines change cluster since the distribution of STDs follows that of EFLs relatively closely. Thus STDs are only of importance near cluster-borders. It can be concluded that deterministic clustering has good stability for uniform sites.

6. Conclusion and recommendations

6.1. Conclusion

Fatigue load differences of 25% for 150 OWTs in 30–40 m water depth showed the importance of site condition variations, making design clustering necessary. A developed probabilistic fatigue load estimation method proved to be suitable for site specific EFL estimation. A verified accuracy of 95% with only seconds of computation time for one simulation case makes the estimation tool ideal for application in clustering optimization, load interpolation, uncertainty studies and preliminary design. Based on EFL estimates, optimization of design clustering gave a reduction of up to 13% of total design loads compared to standardized design. This makes clustering an important approach in support structure cost reduction. By investigation of the marginal gain of adding additional clusters (cf. Fig. 6b), the clustering tool can improve decision-making on the optimal number of clusters of a wind farm in early project phases and is strongly recommended for use in industry. An alternative application of the combined load estimation and clustering tool is wind farm layout design. The layout is typically optimized for power output through wake modeling and cable costs. Having this fast load estimation tool, support structure loads and costs can add a new dimension to layout optimization.

6.2. Recommendations

Two main recommendations for the load estimation tool are: firstly, the verification study should be extended to different structural and environmental settings. More precisely, studies should confirm the accuracy of the estimation method for different support structure eigenfrequencies and in shallow water depths where wind loads have more influence. Ideally, the tool should be validated on experimental data in the future. Secondly, uncertainty in material resistance should be included in the probabilistic

approach making it applicable for reliability-based design. Clustering can then be based on structural reliability instead of (probabilistic) design loads only.

For clustering optimization the following recommendations are made:

- Extend the clustering algorithms for costs optimization.
- Test the local search algorithm for load distributions that feature significant local minima.
- Include load interpolation by formulating the objective function for minimum load difference within clusters. Multi-objective optimization should be implemented to consider EFLs at multiple support structure levels.

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